VALLABHA DATTA VARMA PENMETCHA

Senior Data Engineer | Python • Databricks • GenAI / LLM Applications
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PROFESSIONAL SUMMARY

Senior Data Engineer with 5 years of experience building production-scale data platforms and AI-powered workflows using Python, PySpark, Azure Databricks, and Delta Lake. Built GenAI systems including a RAG chatbot (embeddings, vector search, LLM answering), GPT-4o essay grading, and LLM-based sentiment classification. Designed data models and pipelines serving executive, sales, accounting, marketing, and engineering teams. Delivered Databricks Premium/Unity Catalog migration, cost tracking, and query optimization; skilled in ETL/ELT, CI/CD, code reviews, Agile delivery, and mentoring.

TECHNICAL SKILLS

Languages: Python, PySpark, SQL, Scala, R, Bash

GenAI & Machine Learning: LLMs (GPT-4o), RAG, Embeddings, Vector Search, Prompt Engineering, AI Agents, NLP / Sentiment Classification, Machine Learning, Predictive Modeling, K-means Clustering

Data Platforms: Azure Databricks, Unity Catalog, Delta Lake, Apache Spark, Azure Data Factory, Google BigQuery, GA4, Hadoop, Hive, HBase, Kafka, Oozie

Databases: Microsoft SQL Server (SSMS), Azure Flexible Server, MariaDB, MongoDB

Engineering Practices: ETL/ELT Pipelines, Data Modeling, Query Optimization, Cost Optimization, Data Quality & Validation, Code Reviews, CI/CD

Testing: Unit, Integration, Regression, End-to-End, and Performance Testing

Tools: Git, GitHub, Jenkins, Docker, Jira, Confluence, Agile/Scrum

Visualization: Tableau, Matplotlib, Seaborn, ggplot2, Excel

WORK EXPERIENCE

UWorld India Pvt Ltd — **Hyderabad, India**

Senior Data Engineer I — Jan 2026 – Present

Data Engineer II — Jan 2025 – Dec 2025

- Built an AI-powered chatbot using retrieval-augmented generation (RAG), implementing document ingestion, vectorization, embeddings, vector search, and LLM-based query answering.
- Developed LLM-driven sentiment analysis using prompt engineering to classify and prioritize user comments, directing content improvement efforts.
- Migrated the Databricks workspace to the Premium tier and Unity Catalog, centralizing data governance and access control across catalogs, schemas, and tables.
- Owned the Databricks and Delta Lake redesign aligned with backend SQL Server database separation, enabling seamless scaling for growing users and data volumes with high availability.
- Implemented Databricks cost tracking to monitor compute spend and identify high-cost workloads for optimization.
- Optimized data ingestion, Spark transformations, SQL query performance, and Databricks-to-SQL Server / Azure Flexible Server writes for cost efficiency.
- Integrated GA4, BigQuery, and Azure Databricks to analyze user CTA behavior and deliver data-driven marketing insights.
- Gathered business requirements with stakeholders and designed schemas for new data pipelines.
- Performed code reviews and production deployments, enforcing quality and coding standards; mentored interns and junior engineers on Databricks, Delta Lake, Spark, and data engineering best practices.
- Planned and tracked work in Jira within Agile sprints, coordinating with cross-functional teams for on-time delivery.

Data Engineer I — Oct 2023 – Dec 2024

- Designed and built end-to-end data pipelines in Azure Databricks serving teams across the company — sales, executive, marketing, accounting, and B2B/B2C content — translating each team's requirements into curated, analytics-ready datasets that power their Tableau dashboards and data-driven decisions.
- Developed and productionized an AI-powered automation workflow using GPT-4o to grade user essay submissions with scores and feedback, significantly reducing manual effort and accelerating turnaround time.
- Spearheaded migration of scheduled queries from SQL Server (SSMS) to Azure Databricks, achieving 80% faster execution and improved scalability.
- Developed and maintained scalable ETL pipelines in Azure Databricks to process and transform data for user insights and reporting.
- Designed optimized data models in Delta Lake and MariaDB supporting real-time analytics and batch processing.
- Upgraded Databricks clusters from runtime 9.1 to 12.2 to 14.3 LTS, delivering significant performance and scalability gains.
- Implemented Delta Lake versioning and periodic vacuuming to preserve historical data integrity while reducing storage costs and improving query performance.

- Built data validation and transformation logic in Databricks notebooks and automated workflows with monitoring to ensure pipeline health and timely data delivery.
- Integrated BigQuery with Databricks to prepare marketing funnel data, producing insights that informed subscription and conversion-rate optimization.

Eficens Systems LLC (Client: Target Corporation) **Houston, TX**

Data Engineer Sep 2021 – Aug 2023

- Contributed to rewriting legacy shell scripts as Spark jobs, improving data processing speed by 60%.
- Designed batch processing workflows using Hadoop Streaming, cutting processing time by 40% and reducing infrastructure costs.
- Migrated data processing jobs from Hadoop 2 to Hadoop 3, improving performance and ecosystem compatibility.
- Developed a tombstoning process to identify and suppress outdated forecasts in Hive tables, improving data accuracy and reliability of business insights.
- Applied Hive table optimization techniques, significantly reducing storage costs and improving data extraction efficiency.
- Implemented data quality checks and monitoring to ensure data accuracy and availability for downstream applications.
- Orchestrated pipelines with Oozie coordinators and workflows, built shell scripts for Hive-to-HBase loads, and used Jenkins for CI/CD of data pipelines.
- Managed AD groups, HDFS directories, and Hive/HBase tables, configuring permissions for secure data access.
- Conducted queue sizing analysis and scalability testing, and provided production support to resolve pipeline issues.
- Collaborated with cross-functional teams to integrate data sources for analytics and machine learning initiatives in an Agile/Scrum environment using Jira.

Life Singularity Inc. **Chicago, IL**

Machine Learning Engineer (Internship) Jul 2020 – Sep 2020

- Implemented K-means clustering to identify optimal ranges, improving model performance by 23%.
- Prepared and transformed healthcare claims and diagnosis data in Python (merge, sort, conditional logic), improving Analytics channel performance by 8%.
- Conducted cost-benefit analysis and built dashboards tracking expected-cost changes to inform strategic decisions.

EDUCATION

Master of Science in Data Modeling/Warehousing & Database Administration Sep 2019 – Jun 2021

DePaul University, Chicago, IL | CGPA: 3.85/4

Bachelor of Electrical and Electronics Engineering Aug 2014 – Jun 2018

SASTRA University, Tamil Nadu, India | CGPA: 3.4/4

Relevant Coursework: Regression, Classification, Clustering, Neural Networks, Time Series Forecasting, Advanced Machine Learning, Bayesian Learning, Recommender Systems, Business Intelligence